

Monday Warm Up: Unit 0, Electrostatics I and II

Prof. Jordan C. Hanson

February 3, 2025

1 Memory Bank

- $\vec{F} = q\vec{E}$... Force on a charge q in the presence of an $\vec{E}$ -field.
- $\vec{F} = m\vec{a}$... Newton's 2nd Law.
- $\vec{E}_{\text{plane}} = \frac{\sigma}{2\epsilon_0} \hat{k} = \frac{Q}{2A\epsilon_0} \hat{k}$... The electric field of a plate or disk with charge Q and area A , for distances much smaller than the radius or length.
- **Mass of electron:** 9.11×10^{-31} kg.
- **Charge of electron:** 1.6×10^{-19} C.

2 Electric Fields

1. Consider Fig. 1 below. An inkjet printer shoots microscopic ink droplets through a charging electrode, giving each droplet a charge q . In the region of the deflection plates, there is an electric field $\vec{E}$ pointed upwards. The force on the charged droplets is used to deflect them and draw a shape on the page. Droplets without sufficient charge simply fall to the reservoir.

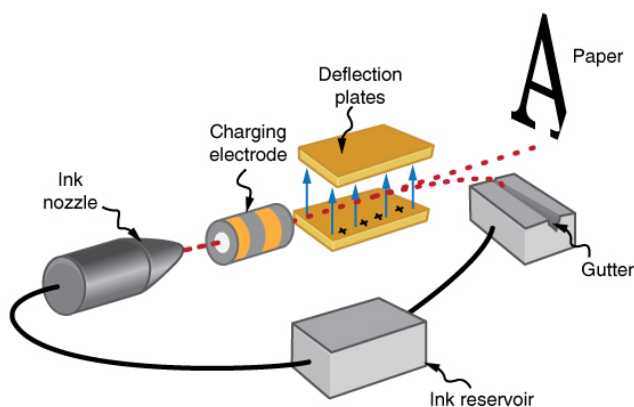


Figure 1: A basic inkjet printer schematic.

2. If the $\vec{E}$ is pointed upwards, and the q on each droplet is positive, the droplets will
 - A: Accelerate downward, if qE exceeds mg
 - B: Accelerate upwards, if qE exceeds mg
 - C: Travel in a straight line, if qE exceeds mg
 - D: Fall to the reservoir



Figure 2: Two charges, q_1 and q_2 are separated by a distance $2d$.

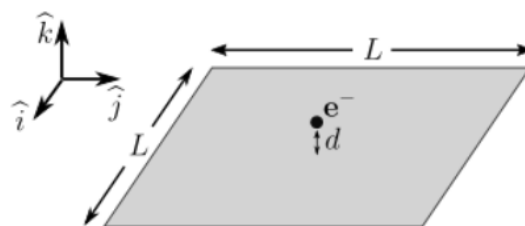


Figure 3: Two charges, q_1 and q_2 are separated by a distance $2d$.

3. Consider Fig. 2. Two charges, q_1 and q_2 are located at $\vec{r}_1 = -d\hat{i}$ and $\vec{r}_2 = d\hat{i}$, respectively. If $q_1 = 1.0 \mu\text{C}$, and $q_2 = 0.5 \mu\text{C}$, and $d = 0.25$ m, where is the electric field zero?
4. Consider Fig. 3. A thin conducting plate with side lengths L is given a charge q . An electron is positioned a distance d above the center of the plate. Assume that $d \ll L$. If $q = 10 \mu\text{C}$, $L = 0.5$ m, and $d = 5$ mm, what is the acceleration of the electron?